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set of functionality and upgrade the features in the future.

In brief, the attempt in R&D is to focus not just on a single multimeter but also on driving customer efficiency by enhancing multimeters' capabilities for a variety of measurements.

Modular platforms combine dynamic and static instruments

Integrating multimeters with other instruments is difficult and extremely challenging, especially in the case of traditional box instrument, as they do not share a common platform.

If each instrument has its own power supply, timing, triggering and synchronisation signals, the only way they can be combined is through external cables as there is no integrated software that can bring them together. This is the reason we need a common, shared modular platform. Modular instruments provide a flexible, organised, software-defined solution for complex integrations.

The latest modular multimeter platforms, such as USB, PCI and PXI, allow the users to make various types of measurements by investing in one particular instrument. But this is possible only if all the combined instruments share a single platform.

In addition to making static DMM-type measurements, the engineers can also make dynamic measurements such as rise time, fall time, FFT and other frequency domain by using such platforms. Since DMM measurements are slower, combining them with faster measurements is always a challenge. But having modularity and a combination of different instruments into one instrument, we can combine the static and dynamic measurements together

in a single platform.

Using certain technology, such as test script processor (TSP), of one particular brand can enhance system control. This technology provides 'smart' instruments the capability of performing distributed processing and control at the instrument level versus a central PC.

Improving user's experience

From my discussions with multimeter vendors, I noticed that most of the vendors are trying to improve user experience rather than improving the specifications of the instrument. The idea behind this is to show how easily a user can use the instrument and how easily it can be integrated with the system. The manufacturers are trying to add more value in terms of the inavailability of that product.

Better-quality displays are very important aspect of improving graphic user interface (GUI). Full-colour, wider graphical displays can be found in some recent multimeters. There are products using organic light-emitting diode (OLED) displays with bright and easier screens for the users to see. But there is still a long way to go for improving customer experience.

Some of the latest models have several built-in features such as back-lighting, charts, histograms, trends and statistics.

Another new feature is the ability to see 'live' measurements simultaneously from multiple modules on a single screen. For example, newer models often include the capability of simultaneously displaying two readings (for example, monitoring voltage and temperature at the same time), measurement trending or even histograms. This makes the GUI more intuitive.

Simplifying user interface is also a priority for developers, so that the tool can be easily handled both by experienced professionals and inexperienced users. As a part of this trend, we find that many multinational brands have been providing multimeters with a selection of local Indian languages such as Hindi, Tamil and Bengali. This is a rather interesting development for a basic product such as a multimeter, where a user is expected to simply read the values. This, I think, would greatly aid in improving user experience.

Safety features

We also find instruments with increased safety, protection level and ruggedness. A safety aspect is seen in recent models, where you have non-contact voltage detection and measurement. This allows the user to know the presence of live wires in the working area even before the tool comes in contact with it. Robustness is important, particularly for handheld multimeters, because such devices are used by people on the move.

What does demand show?

True RMS multimeters are being widely adopted because nowadays people are more aware of the precise measurement requirements. Distributors say they have been observing, over the years, a growing demand for branded multimeters in the market. Users, generally, trust branded multimeters when it comes to the quality, reliability and customer services for the product.

Moreover, several branded multimeter models are now available at a low cost, and hence their demand has increased amongst hobbyists and DIY enthusiasts too. Reducing the voltage range, disabling current meter or reducing the form factor to create pocket-sized instruments are amongst the factors contributing to this reduction in price. ●

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